

Partition, Multiplication, and Order on $\mathbb{Z}$

We have now defined $\mathbb{Z}$ and $+$ such that $(\mathbb{Z}, +)$ forms an abelian group.

Next we wish to show that the intuitive representation of $\mathbb{Z}$ as "positives, zero, and inverses of positives" is correct.

Proposition : $\mathbb{Z} = -\mathbb{N}^+ \uplus \{0\} \uplus \mathbb{N}^+$

Proof:

Consider the set of numbers newly created in the construction of $\mathbb{Z}$:

$$\begin{aligned}\mathbb{Z} \setminus \mathbb{N} &= \{[(a, b)] \in \mathbb{Z} \mid \forall n \in \mathbb{N} : a \neq b + n\} \\ &= \{[(a, b)] \in \mathbb{Z} \mid a <_{\mathbb{N}} b\} \\ &= \{[(a, a + n)] \mid a \in \mathbb{N}, n \in \mathbb{N}^+\} \\ &= \{[(0, n)] \mid n \in \mathbb{N}^+\} \\ &= -\{[(n, 0)] \mid n \in \mathbb{N}^+\} \\ &= -\mathbb{N}^+\end{aligned}$$

Therefore: $\mathbb{Z} = \mathbb{Z} \setminus \mathbb{N} \uplus \mathbb{N} = -\mathbb{N}^+ \uplus \{0\} \uplus \mathbb{N}^+$

The next aim is to define a multiplication $\cdot$ on $\mathbb{Z}$. By difference representation $a - b, c - d \in \mathbb{Z}$ it is clear that we want

$(a - b) \cdot (c - d) = ac + bd - (ad + bc)$ to end up with the usual operation on $\mathbb{Z}$.

Definition:

We define $\cdot : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, $[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)]$. This operation is well defined.

Proof:

Let $[(a, b)] = [(a', b')]$, $[(c, d)] = [(c', d')]$. Then:

$$\begin{aligned}[(a', b')] \cdot [(c', d')] &= [(a'c' + b'd', a'd' + b'c')] \\ &= [(a'c', a'd')] + [(b'd', b'c')] \\ &= [(a'c, a'd)] + [(b'd, b'c)] \\ &= [(a'c, b'c)] + [(b'd, a'd)] \\ &= [(ac, bc)] + [(bd, ad)] \\ &= [(ac + bd, bc + ad)] \\ &= [(a, b)] \cdot [(c, d)]\end{aligned}$$

The structure $(\mathbb{Z}, +, \cdot)$ is an integral domain (= commutative, unitary ring without zero divisors).

The necessary properties follow directly from the definitions. As an example we show that $\mathbb{Z}$ is free of zero divisors.

Proposition : $\forall a, b \in \mathbb{Z} : ab = 0 \implies a = 0 \vee b = 0$

Proof:

Let $[(a, b)], [(c, d)] \in \mathbb{Z}, [(a, b)][(c, d)] = 0$.
 We may additionally assume $[(a, b)] \neq [(0, 0)]$, so $a - b \neq 0$.
 Immediately we receive:

$$\begin{aligned} [(a, b)] \cdot [(c, d)] &= [(ac + bd, ad + bc)] = [(0, 0)] \\ \Rightarrow ac + bd &= ad + bc \\ \Rightarrow ac - bc &= ad - bd \\ \Rightarrow (a - b)c &= (a - b)d \end{aligned}$$

For $k \neq 0$ consider the functions $\varphi_k : \mathbb{Z} \rightarrow \mathbb{Z}, x \mapsto kx$.
 If $k \in \mathbb{N}^+$ we can assume it to be injective. If $k \in -\mathbb{N}^+$ it is a composition of injective maps $\varphi_k = \varphi_{-1} \circ \varphi_{-k}$, and therefore injective as well.

$$\text{Therefore } \varphi_{a-b}(c) = (a - b)c = (a - b)d \varphi_{a-b}(d) \implies c = d \implies [(c, d)] = [(0, 0)]$$

Now comes the ordering on $\mathbb{Z}$.

Definition : For $a, b \in \mathbb{Z}$ we define: $a \leq b \Leftrightarrow b - a \in \mathbb{N}$

Proposition : $\leq$ is a total order.

Proof :

Reflexivity: $\forall a \in \mathbb{Z} : a - a = 0 \in \mathbb{N}$

Antisymmetry: Let $a \leq b, b \leq a$ Then:

$$\begin{aligned} \exists n_1, n_2 \in \mathbb{N} : b - a &= n_1, a - b = n_2 \\ \implies n_1 &= b - a = -n_2 \\ \implies n_1 &= n_2 = 0 \\ \implies a &= b \end{aligned}$$

Transitivity: Let $a \leq b, b \leq c$. Then

$$\begin{aligned} \exists n_1, n_2 \in \mathbb{N} : a + n_1 &= b, b + n_2 = c \\ \Rightarrow \exists n_3 : a + n_3 &= c \\ \Rightarrow a &\leq c \end{aligned}$$

Totality: Let $a, b \in \mathbb{Z}, a \not\leq b$. Then:

$$\begin{aligned} b - a &\notin \mathbb{N} \\ \Rightarrow b - a &\in -\mathbb{N}^+ \\ \Rightarrow a - b &\in \mathbb{N}^+ \\ \Rightarrow b &\leq a \end{aligned}$$

It immediately follows that the partition $\mathbb{Z} = -\mathbb{N}^+ \uplus \{0\} \uplus \mathbb{N}^+$ is the partition in positives, negatives, and zero